Apple's driverless vehicle (Source: https://inhabitat.com)

started out as a frontier to develop robots for mechanical tasks deemed too difficult and unsafe for humans, has fast surpassed its boundaries. Robotics and artificial intelligence (AI) have essentially completely disrupted our lifestyle.

The increasing traffic congestion across major cities has prompted civic authorities and car manufacturers alike to look for possible solutions. Since widening the roads and creating more flyovers wasn't working anymore, the idea of driverless vehicles emerged with automotive, AI and robotics technologies coming together on the same stage.

Autonomous cars will travel across the roads, reading them like grids. These are being programmed to stop at a distance of a few metres under any unpredictable scenario like obstacles or human movement. Automotive and technology companies are coming together for the same.

One may compare the futuristic driverless transport system with the human vascular system. There are about 96,560km (60,000 miles) of blood vessels in our body. These transport blood from one part of the body to another, functioning non-stop 24x7 throughout our lifespan. Yet, in this system of continuous flow, there are no accidents. Also, these occupy a three-dimensional space, unlike our current traffic system that is two-dimensional.

Drawing inspiration from human body, scientists are trying to create transport that can function much like the human vascular system. Concept of gyroscopic vehicles is an excellent example of the same. The future transport will work on a shared basis—much like the veins and arteries in our body, which deliver blood to every tissue and organ in the body but aren't really owned by any particular organ.

You may compare it to the present-day facility of uberPool, with a difference that uberPool takes you around the city choosing the path depending on the drop of each passenger. Driverless cars, on the other hand, would travel on a fixed predefined path. These would stop at definite places and then move on to the next stop only to end where they began.

Are autonomous vehicles road ready?

For the last five years or so we have been reading a lot about driverless car testing in developed countries. It all started with the pilot phase of Google's driverless car in the US. Soon developers realised that the process needs to go through different phases, each phase defining a higher level of accomplishment than the previous ones. This gave the necessary power to the developers of autonomous vehicles to proceed in the right direction.

Let's see how autonomous cars drive on roads: The commuter feeds in the destination point. The AI understands and works on taking up the most suitable route to reach the same. 3D sensors give a complete understanding of the car's environment and path, and thus control actions to be performed by the car. The radar system controls the direction to be taken. The AI stimulates the human perceptual processes for speed, steering wheel balance and brakes. Some automated cars have an override function that allows the human occupant to ride the vehicle.

Are we ready to ride?

There are apprehensions. How can one fully trust technology on the road where the environment is so unpredictable? Automated cars won't be able to manage unforeseen circumstances like the human brain does.

Law makers are concerned about both safety and security. There are grave concerns about the data, and how vulnerable it may make the system to hackers and increase the risk of criminal activities. Data leak may empower enemies to take advantage of numerous citizen lives. EFY